

OOP - 11

Total questions: 8

Worksheet time: 5mins

1.

```
template<typename T>
void f(T obj) {
    cout << "T ";
}

template<>
void f<int>(int obj) {
    cout << "int ";
}

int main() {
    f(3.4);
    f(3);
}
```

a) T int

b) int T

c) T T

d) Няма да се компилира!

2.

```
template<typename T = int>
class Example {
    T el;
public:
    Example(const T& _el) : el(_el) {}
    void ePrint() {
        cout << el;
    }
};

int main() {
    Example<> el(3);
    el.ePrint();
}
```

a) 3

b) Няма да се компилира!

3.

```
template<std::integral T>
class Example {
    T obj;
public:
    Example(const T& _obj);
    void ePrint() const;
};

template <std::integral T>
Example<T>::Example(const T& _obj) : obj(_obj) {}

template <std::integral T>
void Example<T>::ePrint() const {cout << obj;}

int main() {
    Example<double> e(3.14);
    e.print();
}
```

a) 3.14

b) Няма да се компилира!

4.

```
class A {
public:
    A() {
        throw logic_error("Unexpected error!");
        cout << "A() ";
    }
    ~A() {cout << "~A() "; }
};

class B {
public:
    B() { cout << "B() "; }
    ~B() { cout << "~B() "; }
};

void f() {
    B b;
    A a;
}

int main() {
    try {
        f();
    }
    catch (const logic_error& e) {
        cout << e.what();
    }
}
```

a) B()

A()

~A()

~B()

Unexpected error!

c) B()

~B()

Unexpected error!

b) B()

~A()

~B()

Unexpected error!

d) B()

Unexpected error!

5.

```
class A {
public:
    A() {cout << "A() "; }
    ~A() {cout << "~A() "; }
};

class B {
public:
    B() {cout << "B() "; }
    ~B() {cout << "~B() "; }
};

int main() {
    try {
        A a;
        B b;
        throw 'e';
    }
    catch (...) {
        cout << 'e';
    }
}
```

a) A()
B()
~B()
~A()
e

b) A()
B()
e

c) A()
B()
~B()
~A()

d) A()
B()
~A()
~B()
e

6.

```
int main() {
    try {
        throw invalid_argument("error");
    }
    catch (const exception& e) {
        cout << "exception";
    }
    catch (const logic_error& e) {
        cout << "logic_error";
    }
    catch (const invalid_argument& e) {
        cout << "invalid_argument";
    }
}
```

Иерархия: std::exception <- std::logic_error <- std::invalid_argument

a) error

b) exception

c) logic_error

d) invalid_argument

7.

```
class A {
public:
    A() {cout << "A() "; }
    ~A() {
        throw exception();
        cout << "~A() ";
    }
};

void f() {
    A a;
    throw exception();
}

int main() {
    try {
        f();
        cout << "back in main ";
    }
    catch (...) {
        cout << "error caught ";
    }
}
```

a) A()

b) A()

~A()

back in main

error caught

c) A()

back in main

error caught

d) A()

~A()

back in main

8.

```
int main() {
    try {
        throw -1;
    }
    catch (const bad_cast& e) {
        cout << "bad_cast";
    }
    catch (const exception& e) {
        cout << "exception";
    }
    catch (...) {
        cout << "...";
    }
}
```

a) bad_cast

b) exception

c) ...

d) exception ...

Answer Keys

- | | | |
|----------------------------------|---------------------------|-----------------------------|
| 1. a) T int | 2. a) 3 | 3. b) Няма да се компилира! |
| 4. c) B() ~B() Unexpected error! | 5. a) A() B() ~B() ~A() e | 6. b) exception |
| 7. a) A() | 8. c) . . . | |

